12320 Hydrology

Assignment 4: Water Balance for DTU Campus

Hand-in date: Thursday 7 May 2026

In this assignment we will consider the water balance of DTU Campus. You will proceed in the following steps:

1. Characterize the land-use properties of the catchments as a basis for water balance calculations.
2. Compute the radiation balance and reference evapotranspiration according to FAO-56 standard for DTU campus based on meteorological observations.
3. Compute actual evapotranspiration for the different land-use types. In this step you will need to formulate an initial water balance for a soil unit, but you will assume that the influence of infiltration dynamics is negligible.
4. Incorporate infiltration dynamics into the water balance model and evaluate differences in water balance computation.
5. Compute runoff hydrographs at the outlet of the catchment

Please make sure that for any results that you show in this assignment, axes are labelled, line types are clarified using legends and units are included in axis labels, printouts of variables and tables. In addition, the text in any illustrations must be legible for a mid-aged professor without reading aids when printed on an A4 sheet (minimal font size 9). Finally, please make sure to explain symbols shown in equations at least once in your assignment.

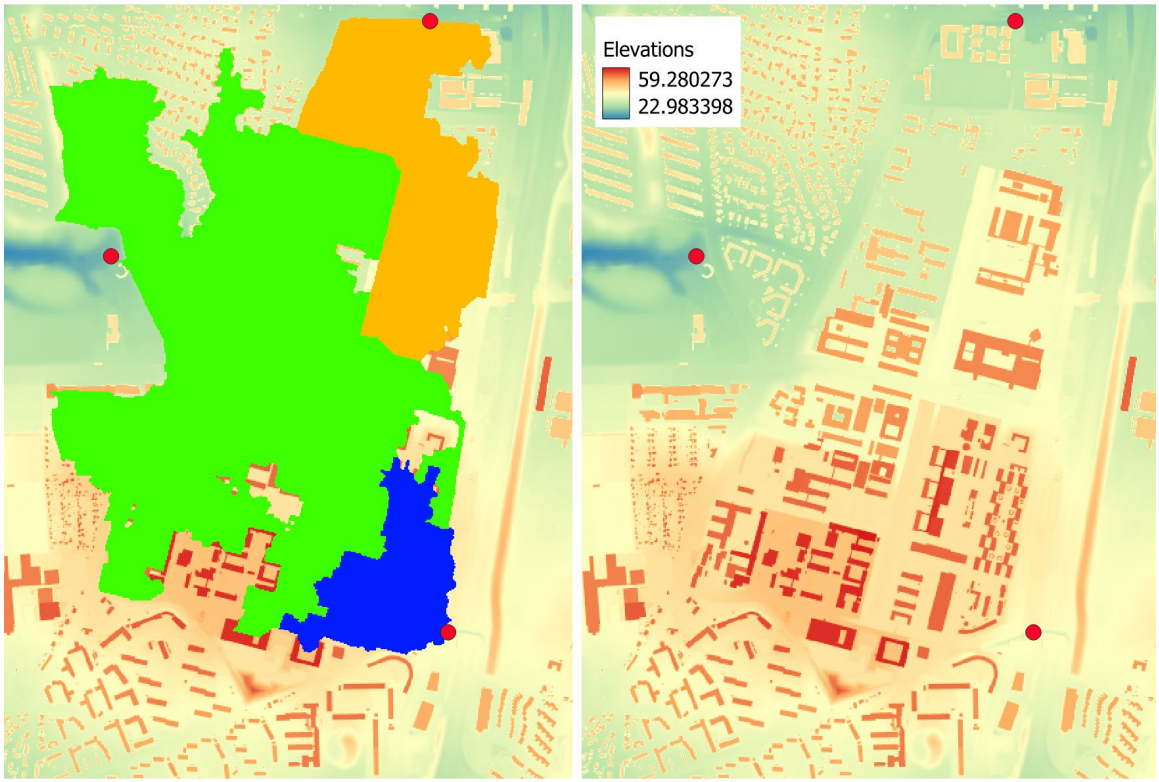
Step 1 - Catchment characterization

The campus can roughly be divided into three catchment areas that deliver surface runoff to different outlets. These are illustrated in the figure below, together with the outlet points and an elevation map in the background.

1) Create a plot showing an overview of the different types of landuses in the catchment
2) Create a table in line with the one below, summarizing the total area of each land use class in each catchment. Which landuse classes do you think are relevant to consider for the water balance calculations?

Land Use Class	Area in catchment green [ha]	Area in catchment orange [ha]	Area in catchment blue [ha]
Bare Land			
Paved areas			
Unpaved road			
Shallow vegetation			
Dense vegetation			
Building			

Total catchment area [ha]			



You are provided with raster files containing landuse characterizations of the campus areas, as well as raster files marking the catchment areas shown in the figure above. The landuse data were obtained from SCALGO. You can find a table showing which number value in the landuse raster corresponds to which type of landuse here (Land cover ID reference):
https://scalgo.com/en-US/scalgo-live-documentation/soil-vegetation-atmosphere-properties/land-cover/mapping

To load the raster data into Python, it is easiest to use the rasterio package (*pip install rasterio*). You can create numpy arrays from the raster files using

```
with rasterio.open('./Land_Cover_Kongens_Lyngby.tif') as src:
    landuse = src.read(1)
```

Step 2 - Radiation balance and reference evapotranspiration



You are provided with meteorological observations from DTU weather station. You can find information about the measured variables (including units) and instrumentation of the station here:
https://weatherdata.construct.dtu.dk/data/

The data have been aggregated to a time resolution of 5 minutes and cleaned from sensor errors. You can therefore use the provided data series as is. Data units remain the same as document on the above web page.

3) Analyze the provided radiation data:
    a. Write down the equation for net radiation and formulate how you will compute each component based on the available data. Consider the albedo for a reference grass surface.
    b. Compute net shortwave radiation, net longwave radiation and net radiation. Plot the time series in hourly and daily aggregation levels. For the hourly aggregation levels, create a zoomed-in plot showing a period of 3 days of your choice.
    Do the patterns look reasonable? Find typical radiation levels in Denmark from public sources and validate your time series.
    Note that LWD data before 2019 have been imputed with typical seasonal patterns.
    c. Compute ground heat flux for your time series. Create a plot showing net radiation and ground heat flux for four winter and four summer days. Interpret the patterns.
4) Compute reference grass evapotranspiration using the FAO-56 Penman Monteith equation in hourly time steps. Analyze evapotranspiration patterns occurring in the course of a day, as well as over the course of a year. Compare your computed ETP values against typical ETP values in Denmark from public sources.

Step 3 – Compute actual evapotranspiration

In this step you will compute actual evapotranspiration for the sparse and dense vegetation classes on the campus. Other than potential evapotranspiration, actual evapotranspiration depends not only on the energy that is available, but also on the available amount of water and on the current state of the vegetation.

We will therefore start by inspecting the observed precipitation time series. In addition, you have been provided with time series of leaf area index for pixels with sparse and dense vegetation on DTU campus. The observations were taken by the NASA Suomi NPP mission.

After inspecting the data, you will use the dual crop coefficient approach (documented in chapters 7, 8 and 9 of the FAO 56 manual) to compute actual evapotranspiration and for land-uses with sparse or dense vegetation. Note that the equations in FAO56 were

derived for daily evapotranspiration, and **you will therefore perform the computation of actual ETP in daily time steps**.

5) Generate a barplot showing monthly precipitation and reference evapotranspiration in the same figure. Interpret the patterns. Compare your precipitation amounts against typical annual precipitation in Denmark as documented in public sources.
6) Plot the LAI series together with a rolling average of air temperature over a period of 30 days. Interpret the results.
7) Generate a daily time series of basal crop coefficients K_{cb} for pixels with natural sparse and dense vegetation. You can assume values at the beginning and end of the growth season are equal and correspond to $K_{cb,ini} = K_{cb,end} = 0.3$; the minimal K_c value for bare soil can be assumed as $K_{c,min} = 0.175$; the vegetation height for sparse vegetation can be assumed constant at 1m, and for dense vegetation at 3m. Start-season in Denmark lasts from Apr 1st to 15th, mid-season through months June, July and August, and the season ends by Nov 1st.
   Plot the derived K_{cb} series.
8) Generate a daily time series of the exposed and wetted soil fraction f_{ew} for both vegetation types based on the basal crop coefficients. Plot the derived series and comment on the patterns.
9) Compute daily time series of soil evaporation coefficients K_e for sparse and dense vegetation. Plot this time series, as well as the daily precipitation, the daily evaporation from the upper soil zone and the depletion of the upper soil zone for the year 2019. Interpret the results.
   Assume that the soil type is silt loam with a water contents of 28% and 15% at field capacity and wilting point, respectively; and that soil evaporation occurs from the upper 10cm of the soil layer only. You can assume that runoff is negligible and that infiltration is unlimited.
10) Compute daily time series of actual evapotranspiration for sparse and dense vegetation. Plot this time series, as well as the daily precipitation, the daily evaporation from the upper soil zone and the depletion of the root zone for the year 2019. Interpret the results.
    Assume the soil type is silt loam as above; that the root zone has a depth of 0.5m for sparse vegetation and 1.6m for dense vegetation; and that 50% of the total available water in the root zone is readily available for transpiration when the plants are not under water stress.

Step 4 – Integrate short-term infiltration dynamics

In this step you will integrate infiltration dynamics into your water balance model, and to what extent these influence the water household.

11) Integrate a dynamic calculation of infiltration dynamics in 5 min time steps into your water balance assessment. Use the Green and Ampt model for this purpose. Use the daily Ks, Kcb and Kr coefficients from the previous step, as well as hourly ET0 to compute hourly actual evapotranspiration. Continue to assume that your soil type is silt loam and choose suitable infiltration parameters.

a. Formulate the soil water balance for daily evapotranspiration considering infiltration dynamics.
b. Compute actual evaporation and infiltration for your time series.
c. Plot precipitation, root zone depletion, actual evapotranspiration, infiltration and runoff as time series in a single figure for the year 2019. Interpret, and compare against the results from step 3.
d. Plot precipitation, runoff and infiltration in 5 minute steps for the two largest rain events in winter 2019 (Dec, Jan, Feb), and the two largest rain events in summer 2019 (Jun, Jul, Aug). Interpret the results

Hint 1: Make a subloop in the daily water balance calculations, where you simulate infiltration dynamics in 5min steps. Consider actual evapotranspiration and root zone depletion when computing I_c and $\Delta\theta$ in the Green-and-Ampt-model. Vice versa, use the Green-and-Ampt model to compute daily infiltration and runoff and use these values as input in the soil water balances for the calculation of actual evapotranspiration. This calculation takes time because you are simulating several years in 5min steps. Consider clipping the simulation period after 2019 to save time.

Hint 2: Use the code below to find all hourly ET0 values for a given day and create a time series in 5min resolution from these. Make sure that all units in your calculation stay consistent.

```
idx_5min = pd.date_range(P_daily.index[ii], periods=288, freq='5min')
ET0_5min = (ET0.reindex(s.index.union(idx_5min)).ffill().reindex(idx_5min))
ET0_5min = ET0_5min*dt/60 #mm/hr to mm per 5min step
```

Step 5 – Runoff hydrograph

As a final step, we will generate a discharge hydrograph for the rain events from question 11d. We will consider only the green catchment. We will assume that the catchment behaves like a natural catchment with a natural stream which, of course, is a major simplification.

12) Use the calculations from question 11 to compute a runoff time series for sparse and dense vegetation for all 4 events.
13) Derive a runoff time series for impervious areas by assuming that the first 1 mm of precipitation in each rain event wettens the surfaces and does not turn into runoff, while 90% of the remaining precipitation turn into runoff.
14) Determine what percentage of the catchment is covered by, respectively, impervious areas, sparse and dense vegetation. You can assume that bare land and unpaved roads behave like sparse vegetation, and that all paved areas and buildings are impervious.
15) Determine which area types contribute most to the total runoff during the considered rain events.
16) Choose a suitable empirical equation for the time of concentration of the catchment and compute the time of concentration. You can assume an average catchment slope S between 0.05 and 0.1 m/m.

17) Determine a discharge hydrograph at the outlet of the catchment using a single linear reservoir. Assume that the time of concentration T_c corresponds to the reservoir constant K. Note that this is a simplification.